NIDAR AirMouse

Team Assignments: Subproblems, Sensor Candidates & Expected Outputs

Purpose: Each team investigates a distinct technical challenge, evaluates suitable sensor configurations, and provides measurable outputs that can be integrated into the final autonomous indoor search-and-rescue pipeline.

Team 1: Indoor Mapping and SLAM

Core question: How can the drone build and continuously update a useful 2D map of an unknown indoor environment?

Problem description

Generate an accurate 2D occupancy-grid map while the drone moves through corridors, junctions and rooms. The map should remain consistent as new areas are explored and should be suitable for live display at the Ground Control Station (GCS).

Sensor candidates to investigate

• RGB-D/depth camera
• 2D LiDAR or short-range ToF/ranging sensor
• Pose estimates from Team 5 as a shared input

Suggested experiments

Compare depth-derived and range-derived mapping; measure map completeness, cell accuracy, update frequency, consistency and memory/computational cost. Investigate loop closure where supported by the selected SLAM approach.

Expected deliverable

A benchmarked 2D mapping/SLAM pipeline, a recommended sensor configuration, and map-quality and runtime measurements.

Team 2: Survivor Detection Under Degraded Conditions

Core question: Can the drone reliably detect survivors when visibility and viewing conditions are poor?

Problem description

Detect human survivors despite low or uneven lighting, partial occlusion, unusual poses, changes in viewing angle and varying distance. The goal is to distinguish genuine detections from false positives and produce reliable detections for downstream localisation.

Sensor candidates to investigate

• RGB camera (including the OAK-D Lite RGB stream, if available)
• Thermal camera such as FLIR Boson, if available
• Event camera such as GenX320, only if its data pipeline is feasible

Suggested experiments

Evaluate precision, recall, F1-score, false positives, missed detections and inference latency across controlled lighting, distance and viewpoint conditions. Test sensor modalities independently before attempting fusion.

Expected deliverable

A tested survivor-detection module, benchmark results under degraded conditions, and an evidence-based recommendation on which sensing modality or combination is justified.

Team 3: Survivor Localisation and Map Tagging

Core question: How can a detected survivor be assigned to the correct grid coordinate on the drone's map?

Problem description

Convert a survivor detection in the camera image into a map-relative position and tag the corresponding grid cell. The location must use the same coordinate frame as the generated map and should account for uncertainty in depth, camera pose and drone localisation.

Sensor candidates to investigate

- RGB-D/depth measurements
- Drone pose and uncertainty from Team 5
- Detection output and confidence from Team 2
- Map and coordinate frame from Team 1

Suggested experiments

Measure map-relative position error, correct grid-cell tagging rate, tag stability and duplicate/false tags. Compare geometric projection approaches and assess how depth and pose uncertainty affect the final location.

Expected deliverable

A survivor localisation and map-tagging module that consumes detection, depth, pose and map inputs and outputs map coordinates/grid cells with confidence information.

Team 4: Obstacle Avoidance and Safe Indoor Navigation

Core question: How can the drone navigate narrow corridors and rooms without colliding with obstacles?

Problem description

Detect nearby walls, corners and obstacles early enough to select safe movements and maintain stable flight in confined indoor spaces. The focus is immediate collision avoidance and local motion safety, rather than deciding the overall exploration order.

Sensor candidates to investigate

- Depth camera
- Short-range LiDAR or ToF sensor
- Optical flow and IMU for motion estimation/stabilisation, where appropriate

Suggested experiments

Compare obstacle detection range, reaction latency, minimum safe clearance, missed obstacles and collision/near-miss rate. Test in representative corridor widths and at different flight speeds.

Expected deliverable

A local obstacle representation, collision-avoidance approach and recommended sensor configuration supported by safety and latency measurements.

Team 5: GPS-Denied Localisation and Drift Reduction

Core question: How can the drone estimate its position and orientation accurately without GPS?

Problem description

Estimate the drone's pose in GPS-denied indoor corridors and rooms while limiting drift, especially in low-texture or repetitive environments. Provide pose estimates in a consistent coordinate frame for mapping, navigation and survivor tagging.

Sensor candidates to investigate

• Camera (monocular, stereo or RGB-D, depending on the available platform)
• IMU
• Optical flow and/or range measurements for comparison

Suggested experiments

Compare visual-only, visual-inertial and range-aided configurations. Measure trajectory/position error and drift under different lighting and corridor conditions; record processing latency and robustness to tracking loss.

Expected deliverable

A benchmarked pose-estimation approach, recommended sensor configuration, pose output with uncertainty where possible, and quantitative drift/error results.

Integration Plan

The teams can investigate their problems in parallel using a shared baseline, recorded data or simulation. Their modules have dependencies, so integration should use agreed interfaces and coordinate frames.

Team	Output to share	Used by
Team 5 — Localisation	Drone pose, timestamps, uncertainty	Teams 1, 3 and 4
Team 1 — Mapping	2D occupancy map, map frame, updates	Teams 3 and 4; GCS integration
Team 2 — Detection	Detection boxes/points, confidence, timestamps	Team 3
Team 4 — Safe navigation	Obstacle data, safe local motion commands/constraints	Mission integration
Team 3 — Tagging	Survivor map coordinates/grid cells, confidence	GCS and final mission pipeline

Shared evaluation rules

For fair comparison, each team should report: (1) accuracy or task-specific quality, (2) processing latency/update rate, (3) compute, memory and power usage where measurable, (4) robustness under degraded conditions, and (5) integration requirements and limitations. Sensor candidates are options to evaluate, not a commitment to install every sensor on the final drone.

Mission constraints to keep in view: GPS/GNSS-denied operation; autonomous mission execution; live mapping and survivor tagging during flight; local communication only; safe navigation; and a maximum 30-minute flight duration under the NIDAR mission brief.